

Let's learn about **satellites!**

Space mission and **satellite** design course

Orbits and Space Environment

Where are we going?

Outer Space, or simply Space, is the expanse that exists outside Earth atmosphere and, more in general, between celestial bodies.



Conventionally, space starts at the **Karman Line**, located at an altitude of 100 km.

What is space made of?

At a first glance, it could seem that space is made of nothing. The reality is a little bit different: space is as **near perfect vacuum**. It contains a low density of particles, as well as electromagnetic rays, neutrinos, dust and cosmic rays.

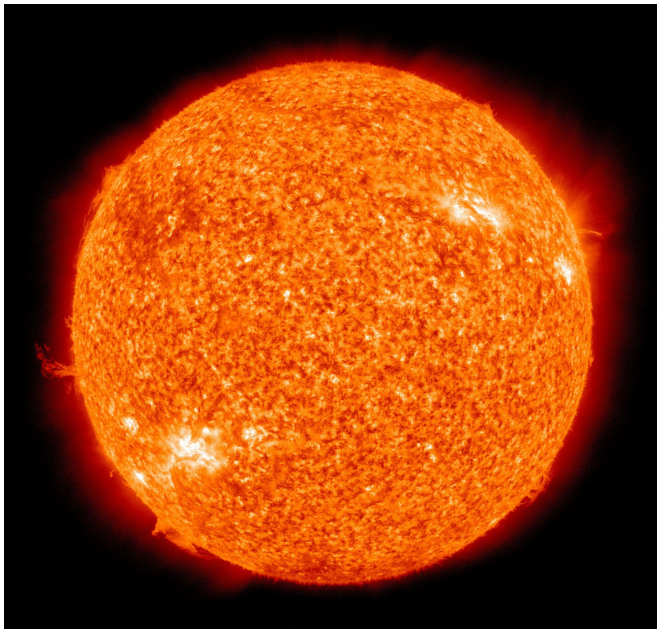
The baseline temperature of the Universe is 2.7 kelvins (-270°C ; -455°F). Cold, isn't it?

Space Environment

Now that we know what space is made of, it is time to discuss why it is so hard to survive in it, both for humans and for spacecrafts. In space, you have to face:

- Solar Radiation
- Plasma Environment
- Residual Atmosphere
- Microgravity
- Near Perfect Vacuum
- Space Debris

Solar Radiation



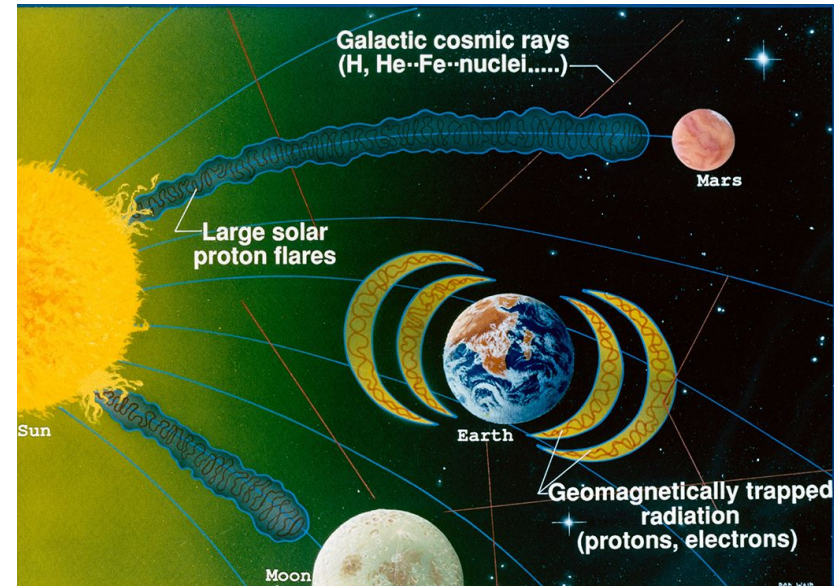
The space in which satellite operate is dominated by the **Sun**.

It is a yellow star whose activities have a 11 years period.

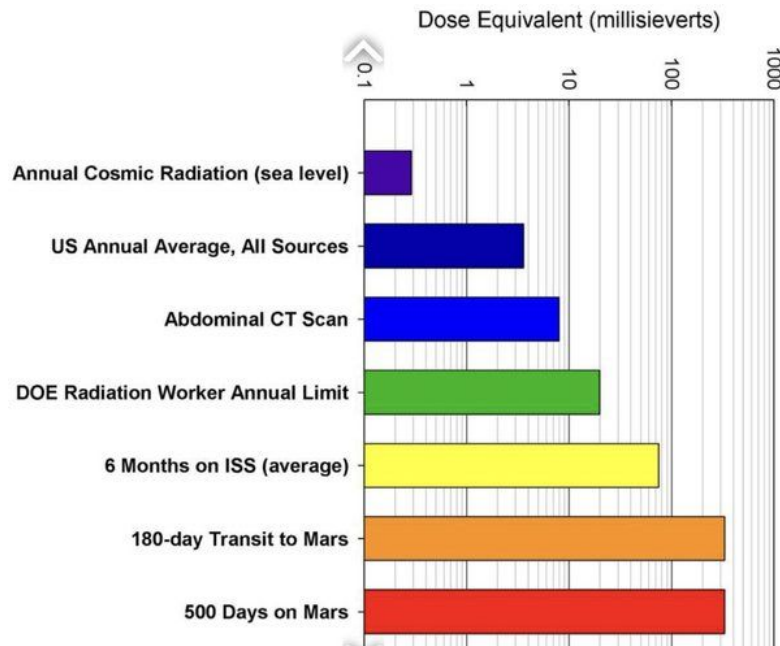
It has a major impact on Thermal Design and on the Power Subsystem, and as we will see soon the particles emitted by the Sun could cause several problem to spacecrafts and humans.

Plasma Environment

The Sun is generating a stream of particles (electrons and protons) that extends up to 100 AU. The Earth magnetic field shields from most of them, but the ones that remain trapped form the Van Allen belts. Also Galactic Cosmic Rays (GCR) are present, contributing to the harsh environment of space.



Plasma Environment



Other than affecting the on-board electronic in many ways, inducing unwanted errors and/or failures, the radiation dose could severely harm human life if too high. The first symptoms of radiation sickness can already be observed on ISS astronauts.

Residual Atmosphere

The density of the atmosphere decreases with altitude, but it is *not equal to zero* and it is relevant especially at lower altitudes. The effect that the atmosphere has on the satellite is called **Drag**: it is a force that opposes to the motion and make the spacecraft fall down to Earth eventually.

We already have seen some of the effect of the Sun, but it affects also the atmosphere density: the atmosphere density is higher if the Sun is more active.

Residual Atmosphere

$$F_{Drag} = -\frac{1}{2} \rho C_D A v^2$$

Drag is proportional to the density ρ , the cross section A and the Drag Coefficient (C_D), that depends on the specific geometry of the satellites. For spacecrafts it usually is around 2 or 2.5.

Even if ρ is extremely low, the orbital velocities are extremely high: the orbital velocity in low Earth orbits is around 8 km/s!

Microgravity

Warning: in space the gravity is still there! A lot of people think that in space we have absence of gravity, but in reality the gravity is almost the same as on Earth. The satellite is in free fall *around* the Earth, and the equilibrium between forces is responsible for the microgravity. In orbit we have gravity around $10^{-6} g$, or one millionth of the gravity we have on Earth. In this condition chemical and physical process are different, as well as human physiology.

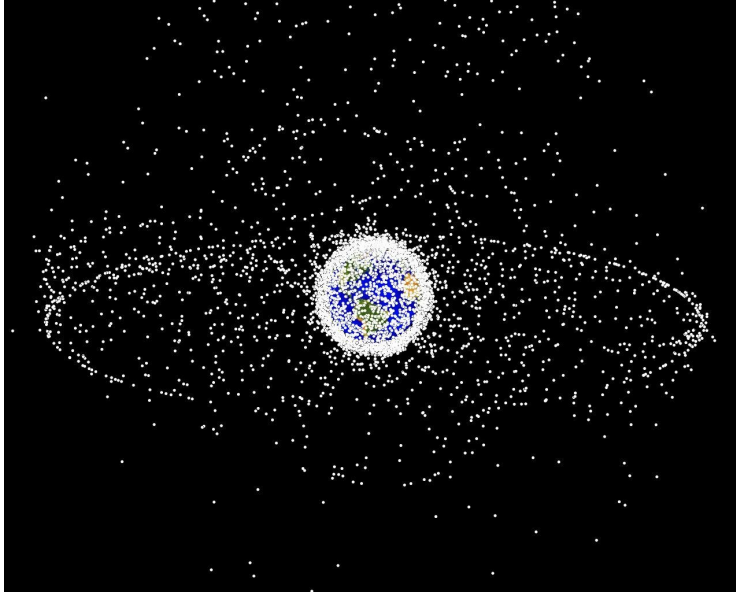
Near Perfect Vacuum

Around Earth pressure is really low (10^{-5} Pa @ 400 km, 10^{-15} Pa @ 36000 km).

If some fluids are trapped inside a material, they “tend to break out. This is a phenomenon known as *outgassing*.

For this reason, not every kind of material can be flown into space, and testing inside vacuum chambers is required before launch.

Space Debris



Around Earth there is a crowded population of natural and artificial objects, like small meteorites, dead satellites, spacecraft parts, fragments and so on.

They are so important that we will have some specific lectures about them.

Orbital Dynamics: Newton's Law

$$F = G \frac{m_1 m_2}{r^2}$$

The gravitational attraction is described by the **Newton's law of universal gravitation**. It states that every body attracts any other body in the Universe with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between their centers.

The two body problem

First of all, we consider only **two body at a time**. In theory, every body in the universe is attracted by any other body in the universe, at any time. Even you, now, are feeling the attraction of Proxima Centauri, Jupiter, the black hole at a center of our galaxy and so on.

Furthermore, we are working in a particular case in which one mass is significantly smaller than the other one: think of a spacecraft compared to a planet! With this assumption, we can find the first fundamental equation of orbital dynamics:

$$\frac{d^2\vec{r}}{dt^2} = -\frac{\mu}{r^2}\hat{r}$$

The two body problem

$$\frac{d^2 \vec{r}}{dt^2} = -\frac{\mu}{r^2} \hat{r}$$

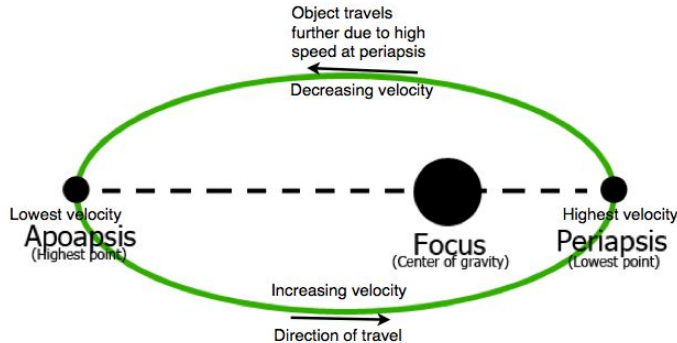
with

$$\mu = GM_{\text{Planet}}$$

μ in this equation depends only on the central body mass: the acceleration of a body in a gravitational field is independent of its mass. This was demonstrated long time ago by **Galileo Galilei** and in his experiment he conducted dropping different masses from the tower of Pisa.

Then, we learn that the gravitational force is a **central force**.

The Keplerian Orbit

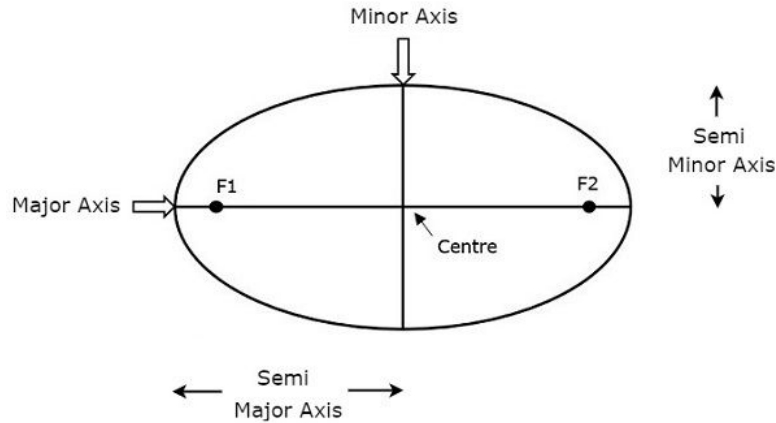


One of the possible solution of the two body problem is the **elliptic orbit**.

For example, the planets in the solar system, and so the Earth, are orbiting around the Sun in elliptic orbits.

The central body is always in one of the two foci. The line joining the central body to the orbiting body sweeps out equal areas in equal amount of time, thus the orbit is faster when the planet is at the periapsis and slower at the apoapsis.

Elliptic Orbit



There are two particularly relevant variables we need to know about an orbit:

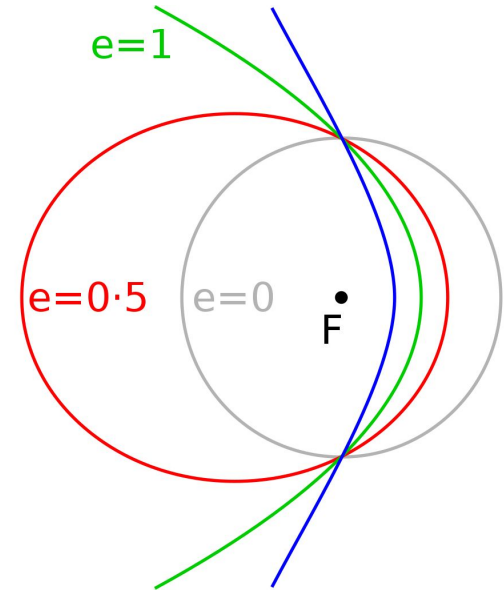
The **semi major axis** referred usually as a is an indication of the dimension of the orbit and it is useful in different calculations.

The **eccentricity**, referred usually as e , is an indication of how much the ellipse is eccentric, from 0 to 1. If it is close to zero, the orbit resemble a circle. The more it grows, the more “squished” the orbit gets. And what happens if it becomes 1 or more?

More orbital shapes

If the eccentricity is equal to zero, the orbit becomes a **circular orbit**. Circular orbits are useful and often used for satellite application.

If $e=1$, we have a **parabola**, and if $e>1$ we have an **hyperbola**. Both have practical application: we will see them together.

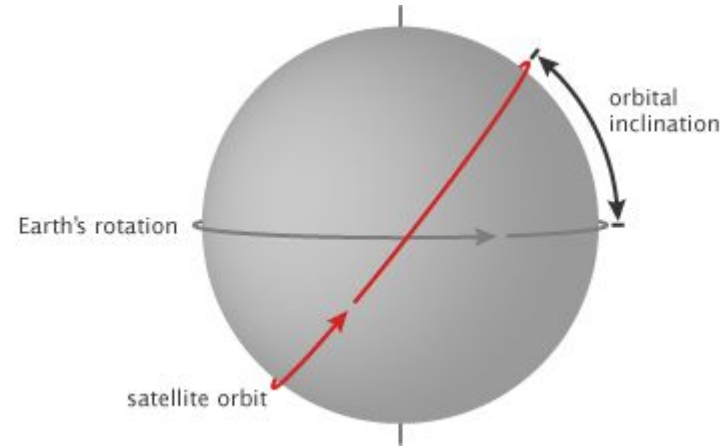


Orbit Inclination

Another important quantity is the **inclination**.

It is usually denoted with the letter i and indicates how much the orbit is tilted with respect to the Earth equator.

If the inclination is 90° , the orbit is passing over the pole and it is referred as a *polar orbit*.



Circular orbit

$$v_{circular} = \sqrt{\frac{\mu}{r}}$$

$$T = 2\pi \sqrt{\frac{r^3}{\mu}}$$

Studying the circular orbit, we can find the **First Cosmic Velocity**, or $v_{circular}$. It is the velocity needed to maintain orbital motion. At zero altitude, it corresponds to 7.9 km/s!

We can also figure out the **Orbital Period**, or the time requested for a celestial body to travel around an orbit.

Exercise - Circular orbit

Given the values on the left, compute:

$$\mu_{Earth} = 398604.3 \frac{km^3}{s^2}$$

$$R_{Earth} = 6371 \text{ km}$$

$$\mu_{moon} = 4904.9 \frac{km^3}{s^2}$$

$$R_{Moon} = 1737.4 \text{ km}$$

$$\mu_{mars} = 42828.4 \frac{km^3}{s^2}$$

$$R_{Mars} = 3389.5 \text{ km}$$

1. The circular velocity and the period of a satellite in a circular orbit around Earth @ 550 km of altitude.
2. The circular velocity and the period of a satellite in a circular orbit around Earth @ 1600 km of altitude.
3. The circular orbit and the period of a satellite in a circular orbit around the Moon @ 550 km of altitude.
4. The circular orbit and the period of a satellite in a circular orbit around mars @ 550 km of altitude.

Solutions - Circular Orbits

1. Velocity = 7.59 km/s; $T = 5730 \text{ s} = 95.5 \text{ min} = 1.59 \text{ hours}$
2. Velocity = 7.07 km/s; $T = 7082 \text{ s} = 118 \text{ min} = 1.97 \text{ hours}$
3. Velocity = 1.464 km/s; $T = 9814 \text{ s} = 163 \text{ min} = 2.72 \text{ hours}$
4. Velocity = 3.30 km/s; $T = 7507 \text{ s} = 125 \text{ min} = 2.09 \text{ hours}$

From this exercise, we can see that the bigger is a planet, the faster a S/C need to fly to maintain a stable orbit; furthermore, the closer a S/C is to a planet the faster it gets.

Parabolic Orbit

$$v_{\text{escape}} = \sqrt{2\frac{\mu}{r}} = \sqrt{2}v_{\text{circular}}$$

If $e=1$, we are in the case of a parabolic orbit. Now the trajectory of the spacecraft is not closed: following a parabolic orbit a satellite can *escape* from the influence of the central body.

This is relevant: if we want for example to go from Earth to Mars, we first need to escape from Earth gravitational attraction.

This is the **Second Cosmic Velocity**, also known as **Escape Velocity**.

Hyperbolic Orbit

$$v_{\infty} = \sqrt{\frac{\mu}{a'}}$$

Even if parabolic orbit allows us to escape from a planet, they have a problem: the velocity of the spacecraft is zero at infinity. For this reason we need hyperbolic orbits. In this case $e > 1$ and the velocity at infinity is greater than zero. The velocity at infinity can be calculated: it is known as the **Third Cosmic Velocity** or Hyperbolic Excess Velocity.

$$v_{\infty}^2 = v^2 - v_{escape}^2$$

If we already know the velocity at which we are flying, we can compute the hyperbolic excess velocity as shown on the left.

Exercise - Parabolic and Hyperbolic Orbits

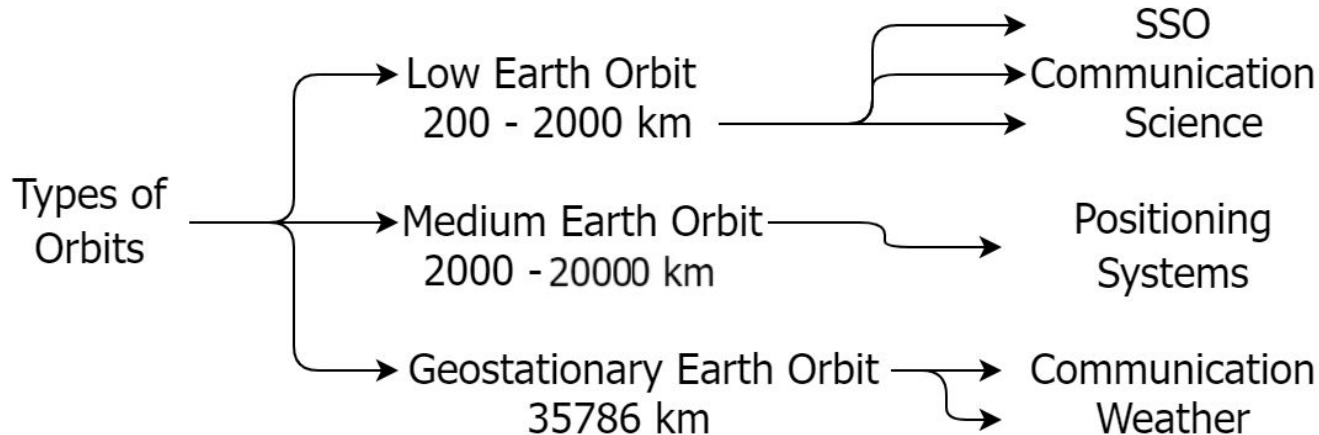
1. Use the same values used in the exercise about circular orbits to compute the escape velocity for every orbit.
2. Your spacecraft is orbiting around Earth with a velocity equal to 7.8 km/s at an altitude of 13800 km .
 - Demonstrate that you are in a Hyperbolic Orbit.
 - Compute the Hyperbolic Excess Velocity.

Solutions - Parabolic and Hyperbolic Orbits

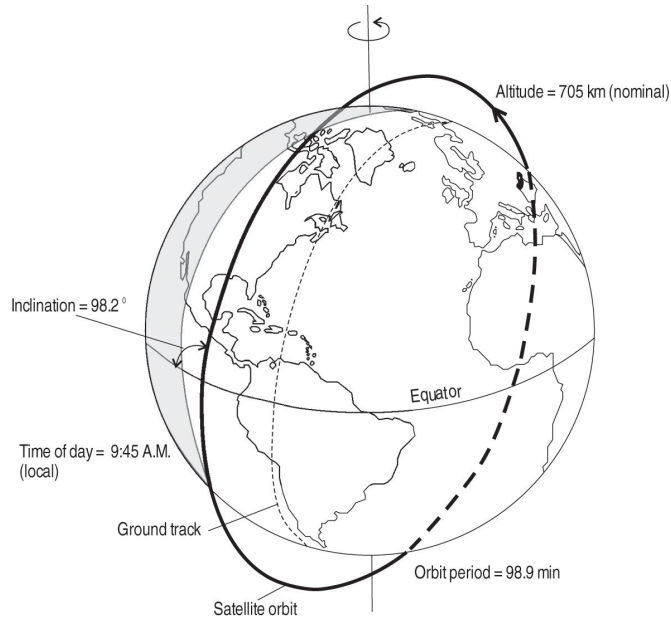
1. The escape velocities are: $V_{\text{esc1}} = 10.73 \text{ km/s}$; $V_{\text{esc2}} = 9.99 \text{ km/s}$; $V_{\text{esc3}} = 2.07$; $V_{\text{esc4}} = 4.66 \text{ km/s}$.
2. First of all, we compute the Escape Velocity in this orbit, corresponding to $V_{\text{esc}} = 6.29 \text{ km/s}$; since our velocity is greater than the escape velocity, we are on an hyperbolic path. Then, we can compute the excess velocity with the equation provided and we obtain $V_{\infty} = 4.61 \text{ km/s}$.

Mission Types and Orbits

The mission types around Earth could be divided into different categories based on the **orbit altitude**. In this way, we can talk about Low Earth Orbit, LEO, Medium Earth Orbits, MEO, and Geostationary Earth Orbit, GEO. Every orbit has its own uses and practical applications.



Low Earth Orbits



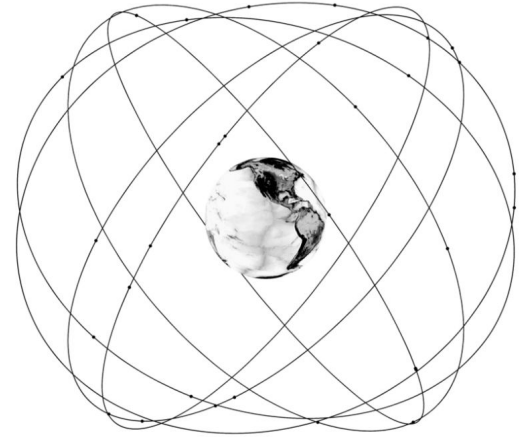
These kind of orbits are the most commonly used. First of all, it is less expensive to send a S/C in Low Earth Orbit. Then, LEOs are particularly useful for a wide range of applications, like Earth Observation, Communication and various science missions.

A particular subset of near-polar orbits, called **Sun Synchronous Orbits**, keeps constant the angle wrt the Sun, ensuring optimal exposure for remote sensing applications.

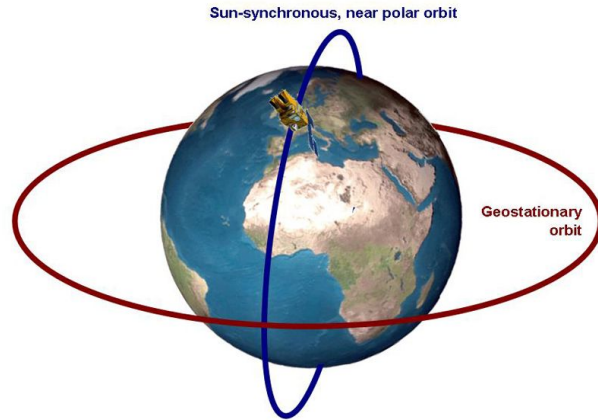
Medium Earth Orbit

MEOs are predictable and reliable orbits often used for **global navigation systems**. GPS, GLONASS, Galileo and BeiDou are all placed in MEOs.

Also specific types of communication satellites make use of MEO, like the Tundra and Molniya . These are **high eccentricity orbits**, so S/Cs are slower (and so further away) when they pass over high latitudes, making them useful for communication in the arctic regions.



Geostationary Orbit

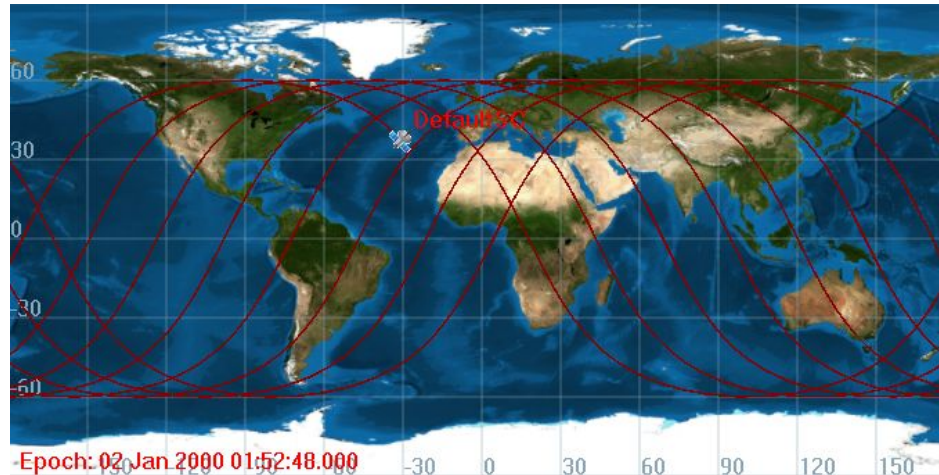


At the specific altitude of 35,786 km, the orbital period is exactly the same as the Earth rotational period, corresponding to **one sidereal day**. This means that a GEO satellite is always on the same point over the Earth, making these kind of satellites particularly useful for communication and weather applications.

Ground Track of a satellite

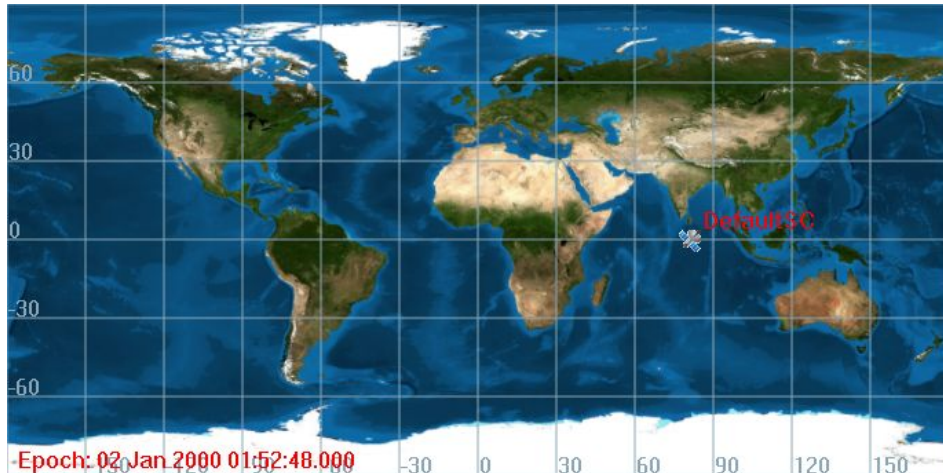
It's extremely relevant to know where a satellite is passing over Earth: you need to know when it passes over the ground stations, or which parts of the Earth you will be able to remote sense. This is usually visualized by the **Ground Track**.

Exercise: try to draw the ground track of a satellite in a GEO orbit.

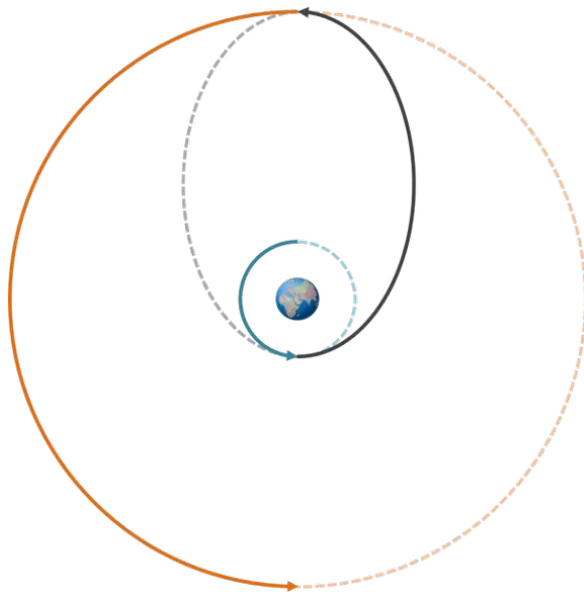


Ground Track of a Satellite

This is the ground track of a GEO satellite! As said before, a satellite in GEO remains on the same point over the Earth. Good work if you guessed it right.



Orbital Transfers

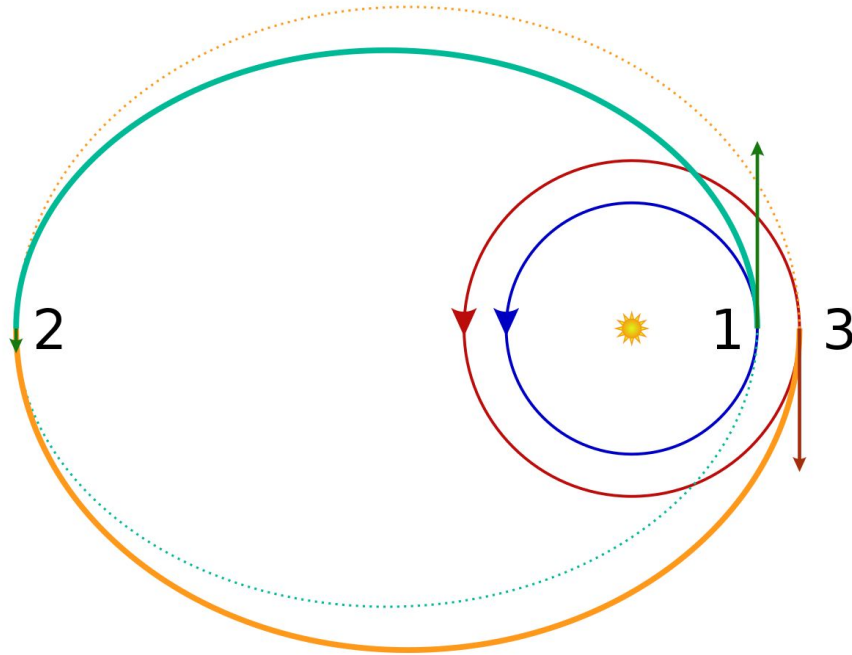


We have seen LEO, MEO and GEO, but how you pass from an orbit to another?

To perform an orbital transfer, you need to use one or more **Transfer Orbits**: orbit that have one point in common with both the starting orbit and the target orbit. In the figure you can see an **Hohmann Transfer**, in which the burst are located at apogee and perigee of the transfer orbit. The Hohmann Transfer is the best possible transfer if:

$$\frac{r_t}{r_s} < 12$$

Bi-elliptic Transfers



In some cases it could be convenient to perform one additional burn and using two transfer orbits. This is the cases for Bi-Elliptic Transfers, that are optimal for:

$$\frac{r_t}{r_s} > 12$$

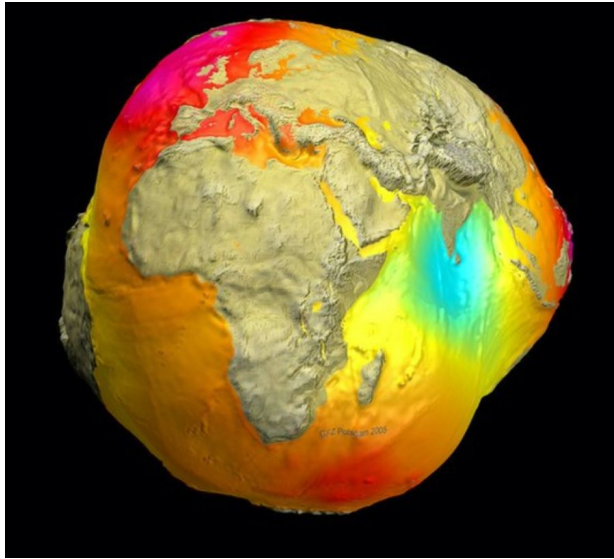
Orbital Perturbations

In reality, all the orbiting bodies are affected by a variety of forces that we call **Orbital Perturbations**.

A real orbit is not a keplerian orbit as we saw earlier, but something really similar modified by the principal perturbations, such as:

- Gravitational potential of the Earth;
- Solar radiation pressure;
- Atmospheric Drag;
- Third body perturbation.

Gravitational Potential of the Earth



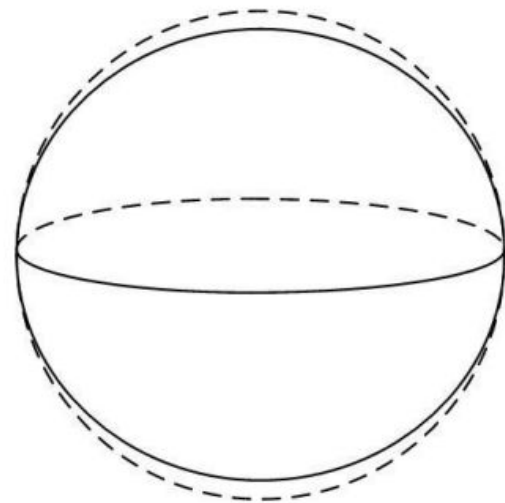
Earth is not a perfect sphere and so the approximation we used to model the keplerian orbit is not correct.

The Earth shape is described as a series of **spherical harmonics**, long mathematical expression that are made of series of coefficients.

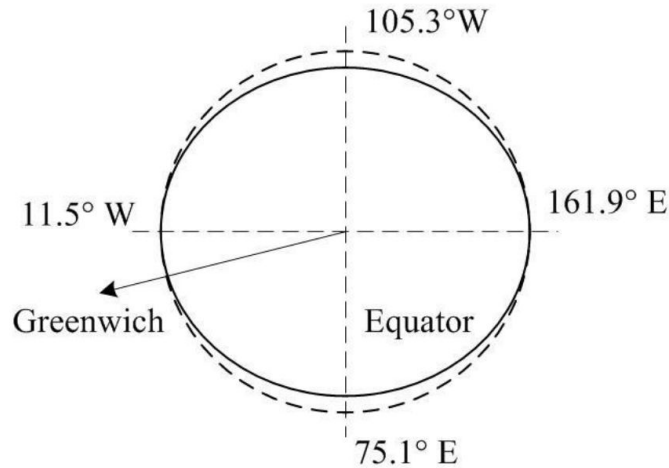
These coefficients, like J_2 and J_{22} , are also used to name the perturbation they affect the most.

Gravitational Potential of the Earth - J2

First of all, Earth is bigger at the equator. The principal effect of this imperfection is to make the orbit tilt around itself (*regression* or *precession*, depending on the inclination), in a similar way of a coin spinning on a table or of a spinning top at the end of its rotation. This effect is called **J2 Perturbation** and it is used to obtain Sun Synchronous Orbits without the need of propellant. At 600 km of altitude we obtain a SSO with an Inclination of 96° .



Gravitational Potential of the Earth - J22



The equator is not a perfect circle, but it resembles more an ellipse.

The **J22 Perturbation** is particularly relevant for equatorial orbits, and thus Geosynchronous orbits. The points at 75.1°E and 105.3°W are points of stable equilibriums, so satellites tend to drift from their positions to this equilibrium points. So every satellite in GEO outside of this point need to perform **Station Keeping Maneuvers** to maintain its position stable.

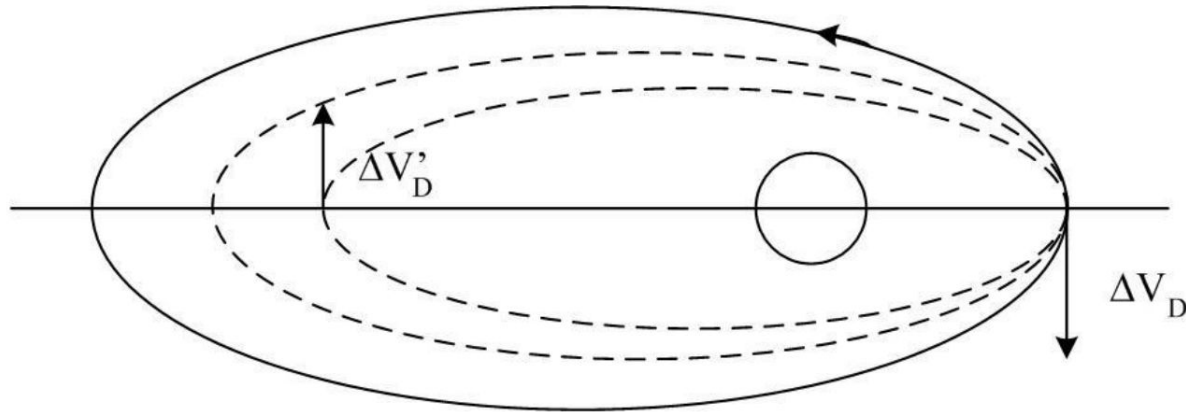
Solar Radiation Pressure

The photons that hit the satellite have a certain amount of momentum. Some of it is transferred to the satellite, changing its momentum. This effect is able to modify the orbital parameters and so has to be accounted. Even if the exerted force is low, **Solar Radiation Pressure** can be used as a mean of propulsion via Solar Sails.



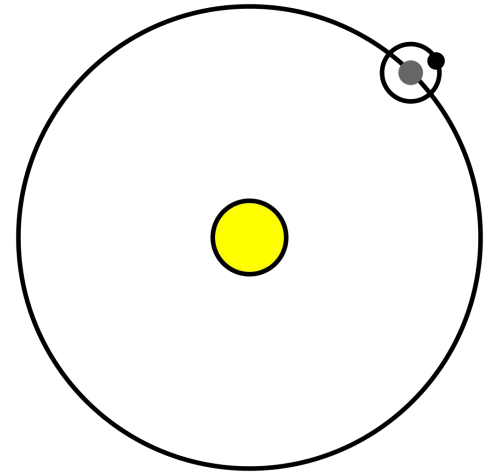
Atmospheric Drag

The **Atmospheric Drag** has two main effects. First, the circularization of the orbit: a highly elliptical orbit will progressively circularize if the perigee is close enough to the Earth surface. this is often used after capture around planet with an atmosphere, like Mars. Second, the decrease of the semi-major axis will eventually cause the fall of the satellite. This could be beneficial at End of Life, but otherwise station keeping maneuvers are requested.



Third Body Perturbations

In our keplerian orbit we only considered two masses: a satellite orbiting around Earth and Earth itself. In reality, there are other bodies that affect the motion of every satellite. For orbits around Earth the more relevant bodies are the Sun, that is gigantic but really far, and the Moon, much smaller but close. Remember the Newton Law: the force of attraction is inversely proportional to the square of the distance and proportional to the mass!



There is much more!

Interplanetary missions, gravity assist, rendez vous maneuvers, constellations, space stations, low thrust maneuvers, orbital refueling, orbital maintenance, rovers, landers, deep space telescopes... We even landed on asteroids and on a comet!

Here we end our overview of Space Environment and Orbital Mechanics... but you can continue study yourself, if you liked what you saw!